

NumPy Reference Documentation

Official Guide to N-Dimensional Arrays, Universal Functions, and Vectorization

1. N-Dimensional Arrays (ndarray)

NumPy provides vectorised multidimensional arrays (`ndarray`) stored in contiguous memory blocks. Memory layouts can be C-contiguous (row-major) or Fortran-contiguous (column-major).

```
import numpy as np

# Create 2D Array and compute matrix operations
a = np.array([[1, 2], [3, 4]], dtype=np.float64)
b = np.array([[5, 6], [7, 8]], dtype=np.float64)

product = np.matmul(a, b)
eigenvalues, eigenvectors = np.linalg.eig(product)
print("Matrix Product:\n", product)
print("Eigenvalues:", eigenvalues)
```

2. Universal Functions & Broadcasting

Broadcasting rules allow numpy operations on arrays with differing dimensions without copying data. Dimensions are aligned right-to-left and checked for equality or dimension size equal to 1.

```
data = np.arange(12).reshape((4, 3))
mean = data.mean(axis=0)
normalized = data - mean # Broadcasts (4,3) minus (3,)
print("Normalized Shape:", normalized.shape)
```